

Fu Zeyuan

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EEE undergraduate who builds LLM applications and AI agents end-to-end - QLoRA fine-tuning, LLM-as-a-judge evaluation, RAG, and on-device GGUF deployment.

EDUCATION

Nanyang Technological University, Singapore

Aug 2023 - Jun 2027

Bachelor of Engineering (Electrical and Electronics Engineering)

- Relevant Modules: (1) Circuit Analysis, (2) Digital Electronics, (3) Semiconductor Fundamentals, (4) Introduction to Data Science and AI, (5) Data Structures and Algorithms in Python, (6) Artificial Intelligence and Data Mining (in progress)

PROJECTS

Personal Project: AI Dungeon Master - Local LLM Agent (LangGraph)

May 2024 - Present

- Built a stateful LangGraph agent (rule-judge -> narrator, checkpointed memory) where the LLM narrates while Python owns game state, dice, and maps - grounding generated text in deterministic game logic; Streamlit UI with voice input and TTS.
- QLoRA 4-bit fine-tuned Qwen2.5-7B-Instruct (Unsloth + TRL, rank-16) on 350 curated instruction dialogues distilled from API transcripts; trains on a consumer RTX 5080 (16 GB) with no cloud GPU spend.
- Built an LLM-as-a-judge evaluation harness (weighted 1-5 rubric: immersion, player agency, rule fidelity) over 50 held-out prompts, comparing the cloud baseline and the fine-tuned local model head-to-head.
- Exported the merged model to GGUF and benchmarked four quantization levels on llama.cpp (Q4_K_M: -69% size, ~2.4x faster generation); added RAG over campaign lore (ChromaDB + MiniLM embeddings); served the model via a self-built OpenAI-compatible FastAPI endpoint.

FYP: Intelligent Control Design and Evaluation for a Mobile Ground Robot to Follow a Human

Aug 2026 - Present

- Modeled a ground robot and its working environment; designed, analyzed, and simulated an intelligent tracking controller combining Fuzzy Logic and Neural Networks for the robot to follow a human, implemented in the Webots simulation environment.
- Evaluated the developed control algorithms through simulation, analyzing tracking performance and robustness.

Multi-Agent Path Finding (MAPF) - EE3180 Design & Innovation Project (Grade A-)

Jan 2026 - May 2026

- Built an interactive Python/Pygame simulator where a centralized GlobalPlanner routes N agents to an exit/parking zone around obstacles, with smart-exit slot assignment and path smoothing.
- Modular control loop separating the planner from the simulation engine: real-time rendering, variable robots, start/reset/speed controls, pause-mode map editor, and per-frame CSV logging.

NTU-EEE Module: Introduction to Data Science and AI

Aug 2024 - Oct 2024

Team Leader of 3

- Built a Python scikit-learn pipeline to clean, train, and evaluate models on the Kaggle cardiovascular disease dataset (70,000 to 68,672 rows after outlier removal).
- Evaluated multiple classifiers and selected the tuned Decision Tree at ~73% test accuracy (GridSearchCV, 10-fold cross-validation); led the team of 3 and kept communication smooth to meet the deadline.

EXPERIENCE

National Service

Feb 2021 - Jan 2023

Electrical Technician

- Installed, maintained, and repaired Signal systems, ensuring 100% operational readiness for mission-critical operations.
- Adapted to high-pressure scenarios (e.g. emergency repairs during field exercises), demonstrating composure and problem-solving skills in resource-limited settings.

Shinryo Corporations Singapore

Jul 2025 - Dec 2025

Electrical Intern

- Prepared and QA'd tender data: equipment schedule data entry + validation/standardization in Excel, using formulas like XLOOKUP/COUNTIF, plus templates/macros to automate repetitive checks and reduce manual errors.
- Supported cost/tender submissions: cable-containment quantity take-offs from drawings and helped draft/update Interconnection Definition Documents (IDDs), coordinating with engineers/BIM to align with Integrated Digital Delivery (IDD) standards.

RELEVANT SKILLS

- Languages: English (fluent), Chinese (Mandarin) (fluent)
- Programming: Python, C++, Linux

- ML/AI: PyTorch, Hugging Face Transformers/TRL, Unsloth (QLoRA), LangGraph/LangChain, RAG (ChromaDB, sentence-transformers), LLM-as-a-judge evaluation, GGUF quantization (llama.cpp), scikit-learn
- Tools: Git/GitHub, FastAPI, Streamlit, Pygame, Webots, Microsoft Office (Excel, Word, PowerPoint)
- Hardware: circuit analysis, soldering, systems maintenance and configuration